

PRIOR FORCE-AWARE VLAs

Vision, language, state, force

Force-aware generative policy

Full action chunk

task motion and contact adjustment fused,
refreshed only at the generative query rate

NO EXPLICIT CORRECTION SUPERVISION

ForceDelta-VLA (ours)

OFFLINE SUPERVISION

Frozen teacher

force-conditioned prediction

missing-force prediction

Explicit force-
correction target

ASYNCHRONOUS DEPLOYMENT

Reference query

cached reference update

100 Hz wrench history

Residual policy

residual corrections compose with the cached
reference, which refreshes independently

COMMANDED POSE AT 100 Hz

distill

